

# Jeff Shen

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## Education

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**H.B.Sc.**, Statistics, Astronomy & Astrophysics, Mathematics 2018-2022  
University of Toronto

## Technical Skills

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**Data:** Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, SQL, NoSQL, R  
**Other:** Rust, BLAS/LAPACK, Docker, bash, vim, git, AWS, SymPy, Astropy, NLTK, L<sup>A</sup>T<sub>E</sub>X, regex

## Research Experience

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| 2021 May - 2021 Aug. | <b>Undergraduate Summer Research Fellowship</b><br><i>Canadian Institute for Theoretical Astrophysics</i><br>Stuff here.                                                                                                                                                                                                                                                                                                                                                                              |
| 2020 Aug. - 2021 May | <b>Undergraduate Research Course</b><br><i>Canadian Institute for Theoretical Astrophysics</i><br>Supervised by <a href="#">Prof. Gwendolyn Eadie</a> and <a href="#">Prof. Norman Murray</a> . Study of mass distribution of the Milky Way in a Bayesian framework. Implemented a hierarchical Bayesian model for modelling the mass of galactic halos using an adaptive gradient-based sampling algorithm, increasing computational efficiency by up to two orders of magnitude over existing code. |
| 2020 May - 2020 Aug. | <b>Summer Undergraduate Research Program</b><br><i>Dunlap Institute for Astronomy and Astrophysics, University of Toronto</i><br>Supervised by <a href="#">Prof. Allison Man</a> . Study of molecular gas in a strongly-lensed galaxy group at $z = 2.9$ . Awarded best poster in student competition. Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication.                                                                    |

## Honours and Awards

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| 2021 May - 2021 Aug. | <b>Undergraduate Summer Research Fellowship, (\$8396 CAD)</b><br>Canadian Institute for Theoretical Astrophysics, University of Toronto    |
| 2020 May - 2020 Aug. | <b>Summer Undergraduate Research Program Award (\$9,500 CAD)</b><br>Dunlap Institute for Astronomy and Astrophysics, University of Toronto |

## Publications

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| 2021 May | <b>Jeff Shen</b> , Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ", submitted to <i>ApJ</i> |
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## Outreach

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| 2019 - Present | <b>Space Science Journalist (Current in Space segment)</b> , <i>The Star Spot</i> podcast<br>Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - present. |
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## Volunteering

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| 2015 - 2018 | <b>Python Workshop Lead</b> , Sir Winston Churchill Secondary, Vancouver, B.C.<br>Coordinated and delivered workshops to prepare students for national programming contest. |
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## Interests

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**Languages:** English (native), French, Chinese, Norwegian (basic)  
**Other:** \*nix customization, computer graphics, 3D modelling, running, baking, reading, guitar